

## Multiple Choice (10 marks)

1. Which of the following is not a security exploit?
  - (a) Eavesdropping
  - (b) Cross-site scripting
  - (c) Authentication
  - (d) SQL Injection
2. \_\_\_\_\_ is the central node of 802.11 wireless operations.
  - (a) WPA
  - (b) Access Point
  - (c) WAP
  - (d) Access Port
3. You are given a message (m) and its OTP encryption (c). Can you compute the OTP key from m and c ?
  - (a) No, I cannot compute the key.
  - (b) Yes, the key is  $k = m \text{ xor } c$ .
  - (c) I can only compute half the bits of the key.
  - (d) Yes, the key is  $k = m \text{ xor } m$ .
4. What was the first buffer overflow attack?
  - (a) Love Bug
  - (b) SQL Slammer
  - (c) Morris Worm
  - (d) Code Red
5. Failed sessions allow brute-force attacks on access credentials. This type of attacks are done in which layer of the OSI model?
  - (a) Physical layer
  - (b) Data-link Layer
  - (c) Session layer
  - (d) Presentation layer

## True / False (10 marks)

*Answer True or False*

6. True or False: The digest created by a hash function is commonly referred to as a Modification Detection Code (MDC).
7. True or False: The Metasploit framework simplifies the exploitation of vulnerabilities through a user-friendly, point-and-click interface.
8. True or False: EXE restarts the STP solver and retries the constraint query indefinitely until a solution is found.
9. True or False: WPA 2 offers a weaker level of security than WEP due to its additional features and complexity.
10. True or False: Encapsulating Security Payload (ESP) is a component of the Secure IP Protocol, providing confidentiality and integrity for IP packets.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the principles and processes of mutation-based fuzzing in cybersecurity, focusing on how it generates diverse inputs by modifying prior inputs. Analyze its effectiveness and potential challenges.
12. Discuss the classic security properties in cybersecurity, identifying which is not traditionally considered one and analyzing its significance in the overall security framework.

*End of Paper*